Exercise J01

Q1. Let P_m stand for ${}^mP_m$. Then the expression $1.\ P_1 + 2.\ P_2 + 3.\ P_3 + \ldots\ldots + n.\ P_n =$

(A) $(n + 1)! - 1$ (B) $(n + 1)! + 1$ (C) $(n + 1)!$ (D) none

Ans. A

Sol. ${}^mP_m = m!$

$1\cdot\underline{|1} + 2\underline{|2} + 3\underline{|3} + \ldots. + n\underline{|n}!$

$$S = \sum_{r=1}^{n} r\cdot\underline{|r} \Rightarrow S = \sum_{r=1}^{n}(r+1-1)\underline{|r}$$

$$S = \sum_{r=1}^{n} \underline{|(r+1)} - \underline{|r}$$

$$S = \underline{|n+1} - 1$$

Q2. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is

(A) 360 (B) 192 (C) 96 (D) 48

Ans. C

Sol. -

Q3. There are two urns. Urn A has 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out at random and then transferred to the other. The number of ways in which this can be done is :-

(A) 3 (B) 36 (C) 66 (D) 108

Ans. D

Sol. -

Q4. There are 10 points in a plans, out of these 6 are collnear. If N is the number of trangles formed by joining these points, then :

(A) $N > 190$ (B) $N \leq 100$ (C) $100 < N \leq 140$ (D) $140 < N \leq 190$

Ans. B

Sol. -

Q5. Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of n is :-

(A) 7 (B) 5 (C) 10 (D) 8

Ans. B

Sol. -

Q6. Let A and B be two sets containing four and two elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is :

(A) 219 (B) 256 (C) 275 (D) 510

Ans. A

Sol. $n(A \times B) = 8$

$^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8 = 219$

Q7. The numbet of integers greater than 6000 that can be formed, using the digits 3, 5, 6, 7, 8, without repetition is :

(A) 216 (B) 192 (C) 120 (D) 72

Ans. B

Sol. -

Q8. A man X has 7 friends, 4 of them are ladies and 3 are men. His wife Y also has 7 friends, 3 of them are ladies and 4 are men. Assume X and Y have no common friends. Then the total number of ways in which X and Y together can throw a party inviting 3 ladies and 3 men, so that 3 friends of each of X and Y are in this party, is

(A) 469 (B) 484 (C) 485 (D) 468

Ans. C

Sol. -

Q9. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. The number of such arrangements is

(A) at least 500 but less than 750 (B) at least 750 but less than 1000

(C) at least 1000 (D) less than 500

Ans. C

Sol. -

Q10. 5 boys & 3 girls are sitting in a row of 8 seats. Number of ways in which they can be seated so that not all the girls sit side by side is:

(A) 36000 (B) 9080 (C) 3960 (D) 11600

Ans. A

Sol. -

Q11. The number of ways in which 6 men and 5 women can dine at a round table if no two women are to sit together is given by

(A) $7! \times 5!$ (B) $6! \times 5!$ (C) $7! \times 6!$ (D) $5! \times 4!$

Ans. B

Sol. -

Q12. The set $S := \{1, 2, 3, ...12\}$ is to be partitioned into three sets A, B, C of equal size. Thus $A \cup B \cup C = S$, $A \cap B = B \cap C = A \cap C = \phi$. The number of ways to partition S is

(A) $\frac{12!}{(4!)^3}$ (B) $\frac{12!}{(4!)^4}$ (C) $\frac{12!}{3!(4!)^3}$ (D) $\frac{12!}{3!(4!)^4}$

Ans. A

Sol. -

Q13. Number of ways in which 9 different toys be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is :

(A) $\frac{(5!)^2}{8}$ (B) $\frac{9!}{2}$ (C) $\frac{9!}{3!(2!)^3}$ (D) None

Ans. C

Sol. -

Q14. The number of words that can be formed by using the letters of the word 'MATHEMATICS' that start as well as end with T is

(A) 80720 (B) 90720 (C) 20860 (D) 37528

Ans. B

Sol.

T										T

Total arrangement is $\frac{9!}{2!.2!} = 90720$

Q15. How many nine digit numbers can be formed using the digits 2, 2, 3, 3, 5, 5, 8, 8, 8 so that the odd digits occupy even positions?

(A) 7560 (B) 180 (C) 16 (D) 60

Ans. D

Sol. -

Q16. How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which not two S are adjacent ?

(A) $6 \times 7 \times {}^8C_4$ (B) $6 \times 8 \times {}^7C_4$ (C) $7 \times {}^6C_4 \times {}^8C_4$ (D) $8 \times {}^6C_4 \times {}^7C_4$

Ans. C

Sol. -

Q17.
Ans. -

Sol. -

Q18. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits
1, 2 and 3 only, is

(A) 55 (B) 66 (C) 77 (D) 88

Ans. C

Sol. -

Q19. How many words can be made with the letters of the word "GENIUS" if each word neither begins with G nor ends in S is:

(A) 24 (B) 240 (C) 480 (D) 504

Ans. D

Sol. -

Q20. The number of positive integers not greater than 100, which are not divisible by 2, 3 or 5 is

(A) 26 (B) 18 (C) 31 (D) none

Ans. A

Sol. Draw a Venn diagram with circles named

A : the number of integers from 1 to 100 divisible by 2 = 50

B : the number of integers from 0 to 100 divisible by 3 = 33

C : the number of integers from 0 to 100 divisible by 5 = 20

$A \cap B$ = 1 to 100 divisible by 2 & 3 = 16

$B \cap C$ = 1 to 100 divisible by 3 & 5 = 6

$A \cap C$ = 1 to 100 divisible by 2 & 5 = 10

$A \cap B \cap C$ = 1 to 100 divisible by 2 & 3 & 5 = 3

Use : $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

$= 50 + 33 + 20 - 16 - 6 - 10 + 3$

$= 106 - 32 \Rightarrow 74$

no. not divisible by 2 & 3 & 5 is = 100 – 74 = 26

Q21. The number of ways in which 8 different flowers can be strung to form a garland so that 4 particulars flowers are never separated is:

(A) $4!.4!$ (B) $\frac{8!}{4!}$ (C) 288 (D) none

Ans. C

Sol. -

Q22. The number of ways in which 6 red roses and 3 white roses (all roses different) can form a garland so that all the white roses come together is

(A) 2170 (B) 2165 (C) 2160 (D) 2155

Ans. C

Sol. -

Q23. A box contains 2 white balls, 3 black balls & 4 red balls. In how many ways can three balls be drawn from the box if atleast one black ball is to be included in draw (the balls of the same colour are different).

(A) 60 (B) 64 (C) 56 (D) none

Ans. B

Sol. Number of ways of selecting 3 balls out of 9 balls without any restriction is ${}^9C_3$.
Number of ways of selecting 3 balls such that no black ball is selected is ${}^6C_3$.
Number of ways such that atleast one black ball is selected is

${}^9C_3 - {}^6C_3 = 84 - 20 = 64$

Q24. Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is :-

(A) 879 (B) 880 (C) 629 (D) 630

Ans. A

Sol. -

Q25. How many divisors of 21600 are divisible by 10 but not by 15?

(A) 10 (B) 30 (C) 40 (D) none

Ans. A

Sol. -

Q26. The sum of the divisors of $2^5 . 3^7 . 5^3 . 7^2$ is

(A) $2^6 . 3^8 . 5^4 . 7^3$ (B) $2^6 . 3^8 . 5^4 . 7^3 - 2 . 3 . 5 . 7$ (C) $2^6 . 3^8 . 5^4 . 7^3 - 1$

(D) none of these

Ans. D

Sol. -

Q27. The number of ways in which the number 27720 can be split into two factors which are co-primes is:

(A) 15 (B) 16 (C) 25 (D) 49

Ans. B

Sol. -

Q28. The streets of a city are arranged like the lines of a chess board. There are m streets running North to South & 'n' streets running East to West. The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is:

(A) $\sqrt{m^2+n^2}$ (B) $\sqrt{(m-1)^2.(n-1)^2}$ (C) $\frac{(m+n)!}{m!.n!}$ (D) $\frac{(m+n-2)!}{m-1)!.(n-1)!}$

Ans. D

Sol. -

Q29. A 3 digit palindrome is a 3 digit number (not starting with zero) which reads the same backwards as forwards. For example 171. The sum of all even 3 digit palindromes, is

(A) 22380 (B) 25700 (C) 22000 (D) 22400

Ans. C

Sol. -

Q30. The sum of all the numbers which can be formed by using the digits 1, 3, 5, 7 all at a time and which have no digit repeated is

(A) $16 \times 4!$ (B) $1111 \times 3!$ (C) $16 \times 1111 \times 3!$ (D) $16 \times 1111 \times 4!$.

Ans. C

Sol. -

Q31. Three digit numbers in which the middle one is a perfect square are formed using the digits 1 to 9 . Their sum is :

(A) 134055 (B) 270540 (C) 170055 (D) none of these

Ans. A

Sol. -

Q32. If chocolates of a particular brand are all identical then the number of ways in which we can choose
6 chocolates out of 8 different brands available in the market is:.

(A) ${}^{13}C_6$ (B) ${}^{13}C_8$ (C) 8^6 (D) none

Ans. A

Sol. -

Q33. Number of positive integral solutions of $x_1 . x_2 . x_3 = 30$ is

(A) 25 (B) 26 (C) 27 (D) 28

Ans. C

Sol. -

Q34. The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is

(A) ${}^{8}C_3$ (B) 21 (C) 3^8 (D) 5

Ans. B

Sol. -

Q35. The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0) is :

(A) 901 (B) 861 (C) 820 (D) 780

Ans. D

Sol. -

Q36. Number of ways in which 3 persons throw a normal die to have a total score of 11 is

(A) 27 (B) 25 (C) 29 (D) 18

Ans. A

Sol. $^{10}C_8 - 3 \cdot {}^4C_2 = 45 - 18 = 27$

Exercise J02

Q1. How many words can be formed by using all the letters of the word 'monday' if each word start with a consonant.

Ans. 480

Sol. -

Q2. In how many ways we can select a committee of 6 persons from 6 boys and 3 girls, if atleast two boys & atleast two girls must be there in the committee.

Ans. 65

Sol. -

Q3. In how many ways playing 11 can be selected from 15 players, if only 6 of these players can bowl and the playing 11 must include atleast 4 bowlers.

Ans. 1170

Sol. -

Q4. एक प्रश्न पत्र में दो भाग है भाग A और भाग B तथा प्रत्येक भाग में 5 प्रश्न हैं। एक छात्र 6 प्रश्नों के उत्तर कितने तरीकों से दे सकता है यदि उसे प्रत्येक भाग में से कम से कम दो प्रश्न अवश्य करने है।

Ans. 200

Sol. -

Q5. How many four digit natural numbers not exceeding the number 4321 can be formed using the digits 1, 2, 3, 4, if repetition is allowed.

Ans. 229

Sol. -

Q6. A committee of 6 is to be chosen from 10 persons with the condition that if a particular person 'A' is chosen, then another particular person B must be chosen.

Ans. 154

Sol. -

Q7. In how many ways a team of 5 can be chosen from 4 girls & 7 boys, if the team has atleast 3 girls.

Ans. 91

Sol. -

Q8. (a) How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if each digit is to be used atmost once.
(b) Find the number of 4 digit positive integers if the product of their digits is divisible by 3.

Ans. (a) 744 (b) 7704

Sol. -

Q9. There are 2 women participating in a chess tournament. Every participant played 2 games with the other participants. The number of games that the men played between themselves exceeded by 66 as compared to the number of games that the men played with the women. Find the number of participants & the total numbers of games played in the tournament.

Ans. 13, 156

Sol. -

Q10. All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, and not divisible by 5 are arranged in the increasing order. Find the $(2004)^{th}$ number in this list.

Ans. -

Sol. -

Q11. An examination paper consists of 12 questions divided into parts A & B.
Part-A contains 7 questions & Part-B contains 5 questions. A candidate is required to attempt 8 questions selecting atleast 3 from each part. In how many maximum ways can the candidate select the questions ?

Ans. 420

Sol. -

Q12. In how many ways can a team of 6 horses be selected out of a stud of 16 , so that there shall always be 3 out of A B C A ' B ' C ' , but never A A ' , B B ' or C C ' together.

Ans. 960

Sol. -

Q13. During a draw of lottery, tickets bearing numbers 1, 2, 3,......, 40, 6 tickets are drawn out & then arranged in the descending order of their numbers. In how many ways, it is possible to have 4th ticket bearing number 25.

Ans. ${}^{24}C_2 . {}^{15}C_3$

Sol. -

Q14. 12 persons are to be seated at a square table, three on each side. 2 persons wish to sit on the north side and two wish to sit on the east side. One other person insists on occupying the middle seat (which may be on any side). Find the number of ways they can be seated.

Ans. 2!3!8!

Sol. -

Q15. (i) If $\frac{1}{9!}+\frac{1}{10!}=\frac{x}{11!}$, find 'x'. (ii) If ${}^{n}P_{5} = 42.\ {}^{n}P_{3}$, find 'n'.

Ans. (i) 121 (ii) 10

Sol. -

Q16. (i) If ${}^{n}C_{3} = {}^{n}C_{5}$, find the value of ${}^{n}C_{2}$. (ii) If ${}^{2n}C_{3} : {}^{n}C_{3} = 11 : 1$, find 'n'.
(iii) If ${}^{n-1}C_{r} : {}^{n}C_{r} : {}^{n+1}C_{r} = 6 : 9 : 13$, find n & r.

Ans. (i) 28 (ii) 6 (iii) n = 12, r = 4

Sol. -

Q17. Find exponent of 3 in 20 !

Ans. 8

Sol. -

Q18. Find number of zeros at the end of 45!.

Ans. 10

Sol. -

Q19. Find the number of words those can be formed by using all letters of the word 'daughter', if all the vowels must not be together.

Ans. 36000

Sol. -

Q20. How many ways fifteen different items may be given to A, B, C such that A gets 3, B gets 5 and remaining goes to C.

Ans. 360360

Sol. -

Q21. A firm of Chartered Accountants in Bombay has to send 10 clerks to 5 different companies, two clerks in each. Two of the companies are in Bombay and the others are outside. Two of the clerks prefer to work in Bombay while three others prefer to work outside. In how many ways can the assignment be made if the preferences are to be satisfied.

Ans. 5400

Sol. -

Q22. A man has 3 friends. In how many ways he can invite one friend everyday for dinner on 6 successive nights so that no friend is invited more than 3 times.

Ans. 510

Sol. -

Q23. Using permutation or otherwise, prove that $\frac{(n^2)!}{(n!)^n}$ is an integer, where n is a positive integer.

Ans. -

Sol. -

Q24. There are 3 cars of different make available to transport 3 girls and 5 boys on a field trip. Each car can hold up to 3 children. Find

(a) the number of ways in which they can be accomodated.

(b) the numbers of ways in which they can be accomodated if 2 or 3 girls are assigned to one of the cars.

In both the cases internal arrangement of children inside the car is considered to be immaterial.

Ans. (a) 1680 (b) 1140

Sol. -

Q25. If all the letters of the word 'again' are arranged in all possible ways & put in dictionary order, what is the 50^{th} word.

Ans. NAAIG

Sol. -

Q26. There are 3 white, 4 blue and 1 red flowers. All of them are taken out one by one and arranged in a row in the order. How many different arrangements are possible (flowers of same colurs are similar).

Ans. 280

Sol. -

Q27. How many different permutations are possible using all the letters of the word MISSISSIPPI, if no two I's are together.

Ans. 7350

Sol. -

Q28. In how many other ways can the letters of the word MULTIPLE be arranged;
(i) without changing the order of the vowels
(ii) keeping the position of each vowel fixed &
(iii) without changing the relative order/position of vowels & consonants.

Ans. (a) 3359, (ii) 59 (iii) 359

Sol. -

Q29. How many arrangements each consisting of 2 vowels & 2 consonants can be made out of the letters of the word **'DEVASTATION'**?

Ans. 1638

Sol. -

Q30. (a) Find the number of words each consisting of 3 consonants & 3 vowels that can be formed from the letters of the word "Circumference". In how many of these c's will be together.
(b) Find the number of 7 lettered words each consisting of 3 vowels and 4 consonants which can be formed using the letters of the word "DIFFERENTIATION".

Ans. (a) 22100, 52 (b) 532770

Sol. -

Q31. Find the number of ways in which the letters of the word 'KUTKUT' can be arranged so that no two alike letters are together.

Ans. 30

Sol. -

Q32. Find ways of selection of atleast one vowel and one consonant from the word TRIPLE

Ans. 45

Sol. -

Q33. In how many ways four men and three women may sit around a round table if all the women are together

Ans. 144

Sol. -

Q34. Find number of divisior of 1980. How many of them are multiple of 11

Ans. 36, 18

Sol. -

Q35. (a) How many divisors are there of the number $x = 21600$. Find also the sum of these divisors.

(b) In how many ways the number 7056 can be resolved as a product of 2 factors.

(c) Find the number of ways in which the number 300300 can be split into 2 factors which are relatively prime.

(d) Find the number of positive integers that are divisors of atleast one of the numbers 10^{10} ; 15^7 ; 18^{11}.

Ans. (a) 72; 78120 (b) 23 (c) 32 (d) 435

Sol. -

Q36. The number of ways in which the number 94864 can be resolved as a product of two factors is____.

Ans. 23

Sol. -

Q37. Determine the number of paths from the origin to the point (9, 9) in the cartesian plane which never pass through (5, 5) in paths consisting only of steps going 1 unit North and 1 unit East.

Ans. 30980

Sol. -

Q38. Find number of negative integral solution of equation $x + y + z = -12$

Ans. 55

Sol. -

Q39. Find number of natural solution of $xyz = 30$

Ans. 27

Sol. -

Q40. A person writes letters to five friends and addresses the corresponding envelops. In how many ways can the letters be placed in the envelops so that

(a) all letters are in the wrong envelops (b) at least three of them are in the wrong envelops.

Ans. (a) 44, (b) 109

Sol. -

Q41. In how many ways it is possible to divide six identical green, six identical blue and six identical red among two persons such that each gets equal number of item.

Ans. 37

Sol. -

Paragraph

There are 8 official 4 non-official members, out of these 12 members a committee of 5 members is to be fromed, then answer the following questions.

\n

Q42. Number of committees consisting of 3 official and 2 non-official members, are

(A) 363 (B) 336 (C) 236 (D) 326

Ans. B

Sol. -

Paragraph

There are 8 official 4 non-official members, out of these 12 members a committee of 5 members is to be fromed, then answer the following questions.

\n

Q43. Number of committees consisting of at least two non-official members, are

(A) 456 (B) 546 (C) 654 (D) 466

Ans. A

Sol. -

Paragraph

There are 8 official 4 non-official members, out of these 12 members a committee of 5 members is to be fromed, then answer the following questions.

\n

Q44. Number of committees in which a particular official member is never included, are

(A) 264 (B) 642 (C) 266 (D) 462

Ans. D

Sol. -

Paragraph

Consider the word W = MISSISSIPPI

Q45. If N denotes the number of different selections of 5 letters from the word W = MISSISSIPPI then N belongs to the set

(A) $\{15, 16, 17, 18, 19\}$ (B) $\{20, 21, 22, 23, 24\}$ (C) $\{25, 26, 27, 28, 29\}$

(D) $\{30, 31, 32, 33, 34\}$

Ans. C

Sol. W = M I S S I S S I P P I

M	I	S	P	
	I	S	P	5 Letters
	I	S		
	I	S		

C–1 2 same + 3 different = $^3C_1 + {}^3C_3 = 4$

C–2 3 same + 2 different = $^2C_1 + {}^3C_2 = 5$

C–3 4 same + 1 different = $^2C_1 + {}^3C_1 = 5$

C–4 3 same + 2 same = $^2C_1 + {}^2C_1 = 4$

C–5 2 same + 2 same + 1 different = $^3C_1 + {}^2C_1 + {}^2C_1 = 7$

$= 25$

Paragraph

Consider the word W = MISSISSIPPI

Q46. Number of ways in which the letters of the word W can be arranged if atleast one vowel is separated from rest of the vowels

(A) $\frac{8! \cdot 161}{4! \cdot 4! \cdot 2!}$ (B) $\frac{8! \cdot 161}{4 \cdot 4! \cdot 2!}$ (C) $\frac{8! \cdot 161}{4! \cdot 2!}$ (D) $\frac{8!}{4! \cdot 2!} \cdot \frac{165}{4!}$

Ans. B

Sol. atleast one vowel is separated from others = total – together

$$= \frac{11!}{4!4!2!} - \frac{8!}{4!2!}$$

$$= \frac{8!}{4!2!}\left[\frac{11 \times 10 \times 9}{4} - 1\right]$$

$$= \frac{8!}{4!2!}\left[\frac{165}{4} - 1\right] \Rightarrow \frac{8! \cdot 161}{4 \cdot 4!2!}$$

Paragraph

Consider the word W = MISSISSIPPI

Q47. If the number of arrangements of the letters of the word W if all the S's and P's are separated is (K) then K equals

(A) $\frac{6}{5}$ (B) 1 (C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Ans. B

Sol. leaving S M – 1 I – 4 P – 2

no. of arrangement $= \frac{7!}{4!2!} = 105$

P together

if P is together no. of arrangement $= \frac{6!}{4!} = 30$

P not together = 105 – 30 = 75

for exp _ P _ M _ P _ I _ I _ I _ I _

No. of ways inserting S = $^8C_4$

for P together = 30

_ P S P _ M _ I _ I _ I _ I _ = $^7C_3$

total arrangements = $^8C_4 \times 75 + 30 \times {}^7C_3$

$= 6300 = k\frac{10!}{4!4!}$

k = 1

Q48. A students has to answer 10 out of 13 questions in an examination. The number of ways in which he can answer, if he must answer atlest three of the first five questions, is

(A) 276 (B) 600 (C) 840 (D) 640

Ans. A

Sol. The possible cases are:

(1 – 5)	(6 – 13)	
3	7	$\rightarrow {}^5C_3 \times {}^8C_7$
4	6	$\rightarrow {}^5C_4 \times {}^8C_6$
5	5	$\rightarrow {}^5C_5 \times {}^8C_5$

Required number of ways

$= {}^5C_3 \times {}^8C_7 + {}^5C_4 \times {}^8C_6 + {}^5C_5 \times {}^8C_5$

$= \frac{5 \times 4}{2} \times 8 + 5 \times \frac{8 \times 7}{2} + \frac{1 \times 8 \times 7 \times 6}{3 \times 2}$

$= 80 + 140 + 56 = 276$

Q49. The number of ways in which 10 students can be divided into three teams, one containing 4 and others 3 each, is

(A) $\frac{10!}{4!3!3!}$ (B) 2100 (C) ${}^{10}C_4 \,.\, {}^5C_3$ (D) $\frac{10!}{6!3!3!} \cdot \frac{1}{2}$

Ans. B, C

Sol. -

Q50. The continued product, 2 . 6 . 10 . 14 ...... to n factors is equal to :

(A) ${}^{2n}C_n$ (B) ${}^{2n}P_n$ (C) (n + 1) (n + 2) (n + 3) ...... (n + n) (D) none

Ans. B, C

Sol. -

Q51. The Number of ways in which five different books to be distributed among 3 persons so that each person gets at least one book, is equal to the number of ways in which

(A) 5 persons are allotted 3 different residential flats so that and each person is alloted at most one flat and no two persons are alloted the same flat.

(B) number of parallelograms (some of which may be overlapping) formed by one set of 6 parallel lines and other set of 5 parallel lines that goes in other direction.

(C) 5 different toys are to be distributed among 3 children, so that each child gets at least one toy.

(D) 3 mathematics professors are assigned five different lecturers to be delivered , so that each professor gets at least one lecturer.

Ans. D, C, B

Sol. (A) We know that, distribution of n distinct objects in to r different boxes, if empty boxes are not allowed or in each box atleast one object is put (n > r).

The number of ways $= r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n + \ldots\ldots (-1)^{r-1}\, {}^rC_{r-1}1$ ....(1)

Hence, n = 5, r = 3

$\therefore$ Number of ways in which the different books be distributed in 3 persons, so each gets atleast one is

$(3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

(B) Number of parallelogram formed by one set of 6 parallel lines and other set of 5 parallel lines

$= {}^6C_2 \times {}^5C_2 = 15 \times 10 = 150$

(C) 5 different toys to be distributed among 3 children, so each gets atleast one

$= (3)^5 - {}^3C_1(2)^5 + {}^3C_2 = 243 - 3 \times 32 + 3 = 150$

(D) 3 mathematics professors are assigned five different lectures, so that each professor get at least one lecture.

$= (3)^5 - {}^3C_1(2)^5 + {}^3C_2 - 1 = 243 - 3 \times 32 + 3 = 150$

Q52. The maximum number of permutations of 2n letters in which there are only a's & b's, taken all at a time is given by:

(A) ${}^{2n}C_n$ (B) $\frac{2}{1}.\frac{6}{2}.\frac{10}{3}.........\frac{4n-6}{n-1}.\frac{4n-2}{n}$

(C) $\frac{n+1}{1}.\frac{n+2}{2}.\frac{n+3}{3}.\frac{n+4}{4}......\frac{2n-1}{n-1}.\frac{2n}{n}$ (D) $\frac{2^n.[1.3.5......(2n-3)(2n-1)}{n!}]$

Ans. A, B, C, D

Sol. -

Q53. The combinatorial coefficient ${}^{n-1}C_p$ denotes

(A) the number of ways in which n things of which p are alike and rest different can be arranged in a circle.

(B) the number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded.

(C) number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty and each box can hold any number of balls.

(D) the number of ways in which (n – 2) white balls and p black balls can be arranged in a line if black balls are separated, balls are all alike except for the colour.

Ans. D, B

Sol. -

Q54. Which of the following statements are correct?

(A) Number of words that can be formed with 6 only of the letters of the word "CENTRIFUGAL" if each word must contain all the vowels is $3 \cdot 7!$

(B) There are 15 balls of which some are white and the rest black. If the number of ways in which the balls can be arranged in a row, is maximum then the number of white balls must be equal to 7 or 8. Assume balls of the same colour to be alike.

(C) There are 12 things, 4 alike of one kind, 5 alike and of another kind and the rest are all different. The total number of combinations is 240.

(D) Number of selections that can be made of 6 letters from the word "COMMITTEE" is 35.

Ans. D, B, A

Sol. -

Q55. Consider all possible permutations of the letters of the word ENDEANOEL
Match the statements / Expression in Column-I with the statements / Expressions in Column-II.

Column-I Column-II

(A) The number of permutations containing the word ENDEA is (P) 5!
(B) The number of permutations in which the letter E occurs in the first and the last position is (Q) $2 \times 5!$
(C) The number of permutations in which none of the letters D, L, N occurs in the last five positions is (R) $7 \times 5!$
(D) The number of permutations in which the letters A, E, O occurs only in odd positions is (S) $21 \times 5!$

(A) (A) S; (B) P; (C) Q; (D) Q (B) (A) Q; (B) S; (C) P; (D) P

(C) (A) P; (B) S; (C) Q; (D) Q (D) (A) S; (B) Q; (C) P; (D) Q

Ans. 5ef2fb4e239ecc768b1680a9

Sol. -

Q56. Column – I Column – II

(A) The total number of selections of fruits which can be made from, 3 alike bananas, 4 alike apples and 2 alike oranges is (P) Greater than 50

(B) If 7 points out of 12 are in the same straight line, then (Q) Greater than 100
the number of triangles formed is
(C) The number of ways of selecting 10 balls from unlimited (R) Greater than 150
number of red, black, white and green balls is
(D) The total number of proper divisors of 38808 is (S) Greater than 200

(A) (A) P (B) PQR (C) PQRS (D) P (B) (A) P (B) R (C) PQRS (D) PQRS

(C) (A) PQR (B) R (C) PQRS (D) P (D) (A) PQ (B) PQRS (C) RS (D) P

Ans. 5ef2eda39fff15766e2df8f5

Sol. -

Q57. Column-I
Column-II
(A) Number of increasing permutations of m symbols are there from the n set
(P) n^m
numbers $\{a_1, a_2, ..., a_n\}$ where the order among the numbers is given by
$a_1 < a_2 < a_3 < ...... a_{n-1} <$ an is
(B) There are m men and n monkeys. Number of ways in which every monkey
(Q) ${}^mC_n$
has a master, if a man can have any number of monkeys
(C) Number of ways in which n red balls and (m – 1) green balls can be arranged
(R) ${}^nC_m$
in a line, so that no two red balls are together, is
(balls of the same colour are alike)
(D) Number of ways in which 'm' different toys can be distributed in 'n' children
(S) m^n
if every child may receive any number of toys, is

(A) (A) R (B) Q (C) S (D) P (B) (A) R (B) S (C) Q (D) P

(C) (A) P (B) S (C) R (D)Q (D) (A) Q (B) P (C) S (D) R

Ans. 5ef2efc69fff15766e2df9f5

Sol. -